Lab 05

Multi-Router Static Routing

A technical portfolio report for Cisco Packet Tracer

AUTHOR	SCOPE	METHOD
Systems Engineering Student | CCNA Candidate	4 routers - 4 LANs - static routes only	Cisco IOS CLI, ARP, ICMP, routing-table analysis

Executive summary

This lab documents the design, implementation, and verification of a four-router IPv4 topology using connected and static routes. The first phase established neighbor reachability; the second phase connected LAN A and LAN D through R1-R3-R4; the final design extended reachability across all four LANs without default routes or dynamic routing protocols. The report also explains the PC9-to-PC7 packet journey, ARP behavior, layer-2 re-encapsulation, TTL changes, and asymmetric return paths.

Lab status: **Static-routing design validated conceptually and partially verified with Packet Tracer captures.** Initial neighbor tests showed 4/5 replies, consistent with first-packet ARP resolution.

1. Objectives

The laboratory objectives were to:

• Build a four-router, four-LAN IPv4 topology in Cisco Packet Tracer.

• Verify directly connected networks and neighbor links before adding remote routes.

• Use specific /24 static routes only; no default routes and no dynamic routing protocol.

• Validate end-to-end connectivity with ping, routing tables, ARP observations, and packet-path reasoning.

• Document the result as a reproducible GitHub portfolio artifact.

2. Topology and addressing

The logical topology is a diamond of four routers. LAN A is attached to R1, LAN B to R3, LAN C to R2, and LAN D to R4. Inter-router links use /30 point-to-point networks.

LAN A 192.168.1.0/24	R1 10.10.12.1 / 10.10.13.1	R3 10.10.13.2 / 10.10.34.1	LAN B 192.168.2.0/24
	R2 10.10.12.2 / 10.10.24.1	R4 10.10.24.2 / 10.10.34.2	LAN D 192.168.4.0/24
		LAN C via R2 192.168.3.0/24	

Figure 1. Logical topology summary. The upper path R1-R3-R4 was used for the initial LAN A to LAN D validation.

Device	Role / LAN	Interface or link	Addressing
R1	LAN A gateway	LAN A; R1-R2; R1-R3	192.168.1.1/24; 10.10.12.1/30; 10.10.13.1/30
R2	LAN C gateway	R1-R2; R2-R4	192.168.3.1/24; 10.10.12.2/30; 10.10.24.1/30
R3	LAN B gateway	R1-R3; R3-R4	192.168.2.1/24; 10.10.13.2/30; 10.10.34.1/30
R4	LAN D gateway	R2-R4; R3-R4	192.168.4.1/24; 10.10.24.2/30; 10.10.34.2/30
PC9	LAN A host	Default gateway R1	192.168.1.24/24; GW 192.168.1.1
PC7	LAN D host	Default gateway R4	192.168.4.39/24; GW 192.168.4.1

3. IOS procedures

The configuration sequence followed a dependency-aware workflow: verify interfaces and directly connected routes, test neighbors, add static routes, save the configuration, and then test remote LANs.

Baseline verification

```
show ip interface brief
show ip route
show ip arp
ping 10.10.13.2
ping 10.10.12.2
ping 10.10.24.2
ping 10.10.34.2
```

Static-route syntax

```
ip route <destination-network> <mask> <next-hop>
copy running-config startup-config
```

Implemented path for LAN A to LAN D

```
R1: ip route 192.168.4.0 255.255.255.0 10.10.13.2
R3: ip route 192.168.4.0 255.255.255.0 10.10.34.2
R4: ip route 192.168.1.0 255.255.255.0 10.10.34.1
R3: ip route 192.168.1.0 255.255.255.0 10.10.13.1
```

Design correction

The proposed next hop 10.10.32.1 for R4 toward LAN B was rejected because the topology contains no 10.10.32.0/30 link. The valid R4-to-R3 link is 10.10.34.0/30, so R3 is reached through 10.10.34.1.

4. Routing-table view: C / L / S

Each router should show connected (C), local host (L), and static (S) entries. The tables below summarize the expected forwarding knowledge after the complete static-routing phase.

Router	Connected / local knowledge	Static routes required for remote LANs
R1	C 192.168.1.0/24; C 10.10.12.0/30; C 10.10.13.0/30; L addresses	S 192.168.2.0/24 via 10.10.13.2; S 192.168.3.0/24 via 10.10.12.2; S 192.168.4.0/24 via 10.10.13.2
R2	C 192.168.3.0/24; C 10.10.12.0/30; C 10.10.24.0/30; L addresses	S 192.168.1.0/24 via 10.10.12.1; S 192.168.2.0/24 via 10.10.24.2; S 192.168.4.0/24 via 10.10.24.2
R3	C 192.168.2.0/24; C 10.10.13.0/30; C 10.10.34.0/30; L addresses	S 192.168.1.0/24 via 10.10.13.1; S 192.168.3.0/24 via 10.10.13.1; S 192.168.4.0/24 via 10.10.34.2
R4	C 192.168.4.0/24; C 10.10.24.0/30; C 10.10.34.0/30; L addresses	S 192.168.1.0/24 via 10.10.34.1; S 192.168.2.0/24 via 10.10.34.1; S 192.168.3.0/24 via 10.10.24.1

Note: equivalent static paths can be valid if the next hop is reachable and the full return path is installed. The table reflects the selected design and the route decisions discussed in the lab.

5. Verification results

Neighbor tests captured in Packet Tracer returned 4/5 replies (80%) on the four inter-router links. The first dot in the IOS pattern .!!!! represents the timed-out first echo; subsequent replies succeeded. This is compatible with first-packet ARP resolution, although ARP must be checked directly with show ip arp to prove the cause.

Test	Observed result	Interpretation
R1 -> R3 (10.10.13.2)	4/5 replies; 80%	Directly connected neighbor reachable; first packet likely preceded ARP completion.
R1 -> R2 (10.10.12.2)	4/5 replies; 80%	Directly connected neighbor reachable.
R2 -> R4 (10.10.24.2)	4/5 replies; 80%	Directly connected neighbor reachable.
R3 -> R4 (10.10.34.2)	4/5 replies; 80%	Directly connected neighbor reachable.
PC9 -> PC7	3/4 replies; 25% loss in captured test	End-to-end path existed; repeat after ARP caches stabilize to confirm 100%.

Recommended repeat verification: clear or inspect ARP state, run show ip arp, then repeat the same pings and record whether the first-packet loss disappears.

6. Packet journey: PC9 to PC7

PC9 (192.168.1.24) determines that 192.168.4.39 is remote and therefore resolves the MAC address of its default gateway, 192.168.1.1, not the MAC of PC7. R1 receives the frame, removes the Ethernet header, consults its routing table, decrements TTL, and creates a new Ethernet frame toward R3 using next hop 10.10.13.2. R3 repeats the process and forwards toward R4 through 10.10.34.2. R4 sees 192.168.4.0/24 as directly connected, uses ARP if needed to resolve PC7, and sends the final frame as unicast after the ARP exchange.

Segment	IPv4 source -> destination	Layer-2 / forwarding action
PC9 -> R1	192.168.1.24 -> 192.168.4.39	PC9 ARPs for 192.168.1.1; frame destination is R1 MAC.
R1 -> R3	same IP endpoints	New Ethernet frame; next hop 10.10.13.2; TTL decreases by 1.
R3 -> R4	same IP endpoints	New Ethernet frame; next hop 10.10.34.2; TTL decreases by 1.
R4 -> PC7	same IP endpoints	R4 ARPs for 192.168.4.39 if needed; final unicast delivery.

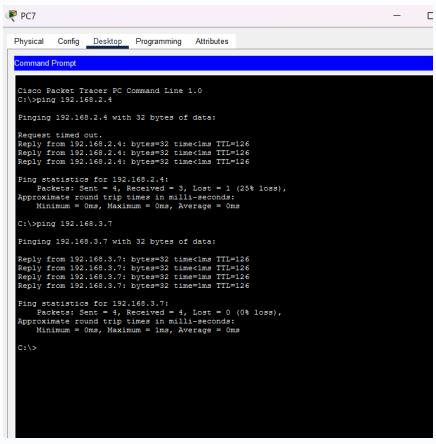
7. Asymmetric routing and ARP findings

The forward path can differ from the return path. For example, PC9 to PC7 may travel PC9 -> R1 -> R3 -> R4 -> PC7, while the ICMP Echo Reply may return PC7 -> R4 -> R3 -> R1 -> PC9. This is asymmetric routing. It remains valid here because every router has a route for the reverse destination and no stateful security policy is restricting the return direction.

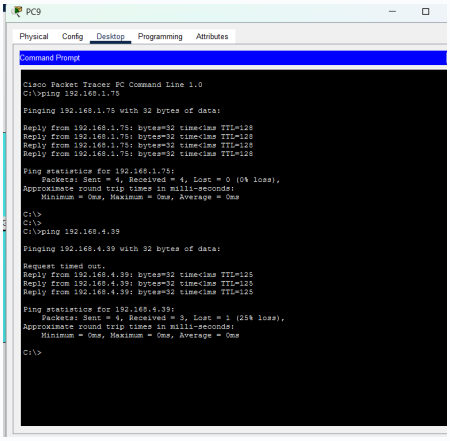
ARP distinction: ARP resolves an IP-to-MAC relationship on a local link. A switch learns MAC-to-port relationships from source MAC addresses. A router does not broadcast the packet to all end hosts; it broadcasts only an ARP Request when it needs the local destination MAC.

8. Evidence captured

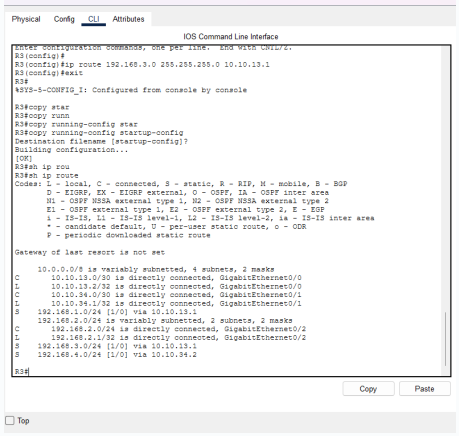
The following Packet Tracer captures were supplied with the lab and are included as visual evidence. Captions identify their role without claiming details that are not legible in every frame.



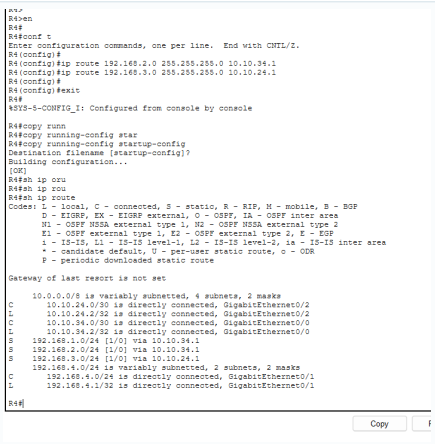
Evidence 1: Packet Tracer capture



Evidence 2: Packet Tracer capture



Evidence 3: Packet Tracer capture



```
R4>en
R4#
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#
R4(config)#ip route 192.168.2.0 255.255.255.0 10.10.34.1
R4(config)#ip route 192.168.3.0 255.255.255.0 10.10.24.1
R4(config)#
R4(config)#exit
R4#
%SYS-5-CONFIG_I: Configured from console by console

R4#copy runn
R4#copy running-config star
R4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R4#sh ip oru
R4#sh ip rou
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

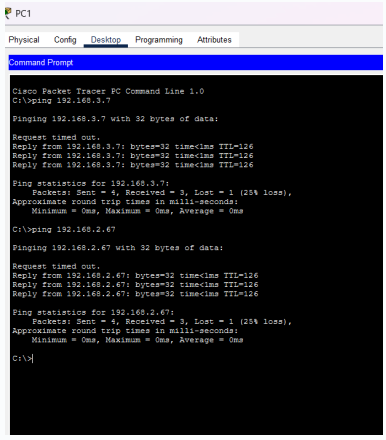
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.24.0/30 is directly connected, GigabitEthernet0/2
L       10.10.24.2/32 is directly connected, GigabitEthernet0/2
C       10.10.34.0/30 is directly connected, GigabitEthernet0/0
L       10.10.34.2/32 is directly connected, GigabitEthernet0/0
S    192.168.1.0/24 [1/0] via 10.10.34.1
S    192.168.2.0/24 [1/0] via 10.10.34.1
S    192.168.3.0/24 [1/0] via 10.10.24.1
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/1
L       192.168.4.1/32 is directly connected, GigabitEthernet0/1

R4#
```

Copy

Evidence 4: Packet Tracer capture



PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.7

Pinging 192.168.3.7 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.7: bytes=32 time<1ms TTL=126
Reply from 192.168.3.7: bytes=32 time<1ms TTL=126
Reply from 192.168.3.7: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.7:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.67

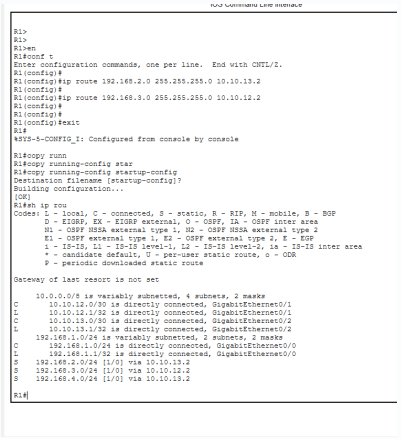
Pinging 192.168.2.67 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.67: bytes=32 time<1ms TTL=126
Reply from 192.168.2.67: bytes=32 time<1ms TTL=126
Reply from 192.168.2.67: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Evidence 5: Packet Tracer capture



```
R1>
R1>
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ip route 192.168.2.0 255.255.255.0 10.10.13.2
R1(config)#
R1(config)#ip route 192.168.3.0 255.255.255.0 10.10.12.2
R1(config)#
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

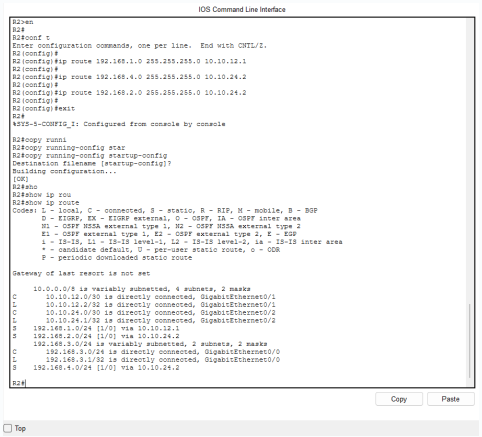
R1#copy runn
R1#copy running-config star
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
L       10.10.12.1/32 is directly connected, GigabitEthernet0/1
C       10.10.13.0/30 is directly connected, GigabitEthernet0/2
L       10.10.13.1/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.10.13.2
S    192.168.3.0/24 [1/0] via 10.10.12.2
S    192.168.4.0/24 [1/0] via 10.10.13.2

R1#
```

Evidence 6: Packet Tracer capture



IOS Command Line Interface

```
R2>en
R2#
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#ip route 192.168.1.0 255.255.255.0 10.10.12.1
R2(config)#
R2(config)#ip route 192.168.4.0 255.255.255.0 10.10.24.2
R2(config)#
R2(config)#ip route 192.168.2.0 255.255.255.0 10.10.24.2
R2(config)#
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy runni
R2#copy running-config star
R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#sho
R2#show ip rou
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

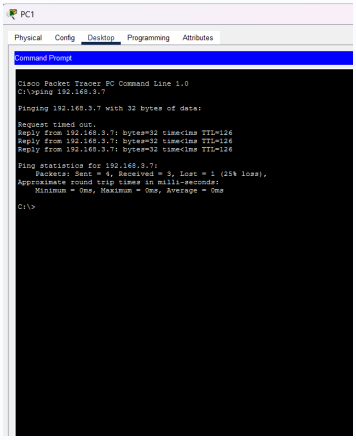
     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
L       10.10.12.2/32 is directly connected, GigabitEthernet0/1
C       10.10.24.0/30 is directly connected, GigabitEthernet0/2
L       10.10.24.1/32 is directly connected, GigabitEthernet0/2
S    192.168.1.0/24 [1/0] via 10.10.12.1
S    192.168.2.0/24 [1/0] via 10.10.24.2
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
S    192.168.4.0/24 [1/0] via 10.10.24.2

R2#
```

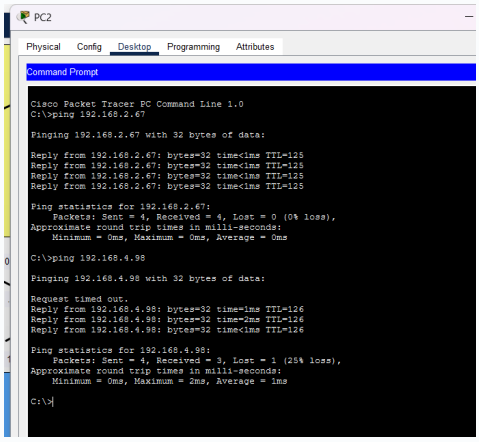
Copy Paste

Top

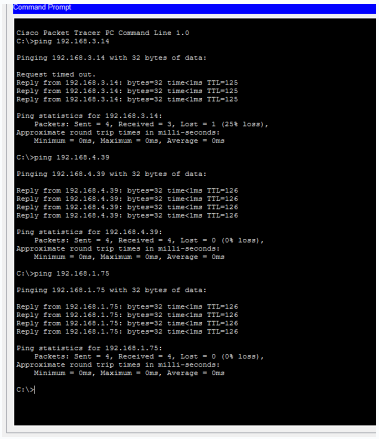
Evidence 7: Packet Tracer capture



Evidence 8: Packet Tracer capture



Evidence 9: Packet Tracer capture



Evidence 10: Packet Tracer capture

9. Conclusions and lessons learned

This lab connected the mechanics of routing to observable behavior in Packet Tracer. The most important result was not simply that a ping succeeded; it was understanding why the first packet can be delayed by ARP, why each router creates a new layer-2 frame, why TTL changes at every routed hop, and why successful communication requires a complete path in both directions.

The route review also reinforced a practical design rule: a next hop must exist on a directly reachable subnet. The invalid 10.10.32.1 proposal was corrected to 10.10.34.1 because the actual R3-R4 transit network is 10.10.34.0/30.

For a GitHub repository, the recommended companion files are the Packet Tracer .pkt file, four sanitized running configurations, a README with the addressing plan, and the original evidence images. The PDF provides the narrative record and can be linked from the lab folder.

Final checklist

✓ Four LANs identified and addressed with /24 networks.

✓ Four point-to-point links identified and addressed with /30 networks.

✓ Static-route design uses specific routes and no default route.

✓ Return paths and asymmetric routing are explicitly analyzed.

✓ ARP, MAC learning, encapsulation, TTL, and ICMP verification are documented.

✓ Evidence captures are included for repository publication.